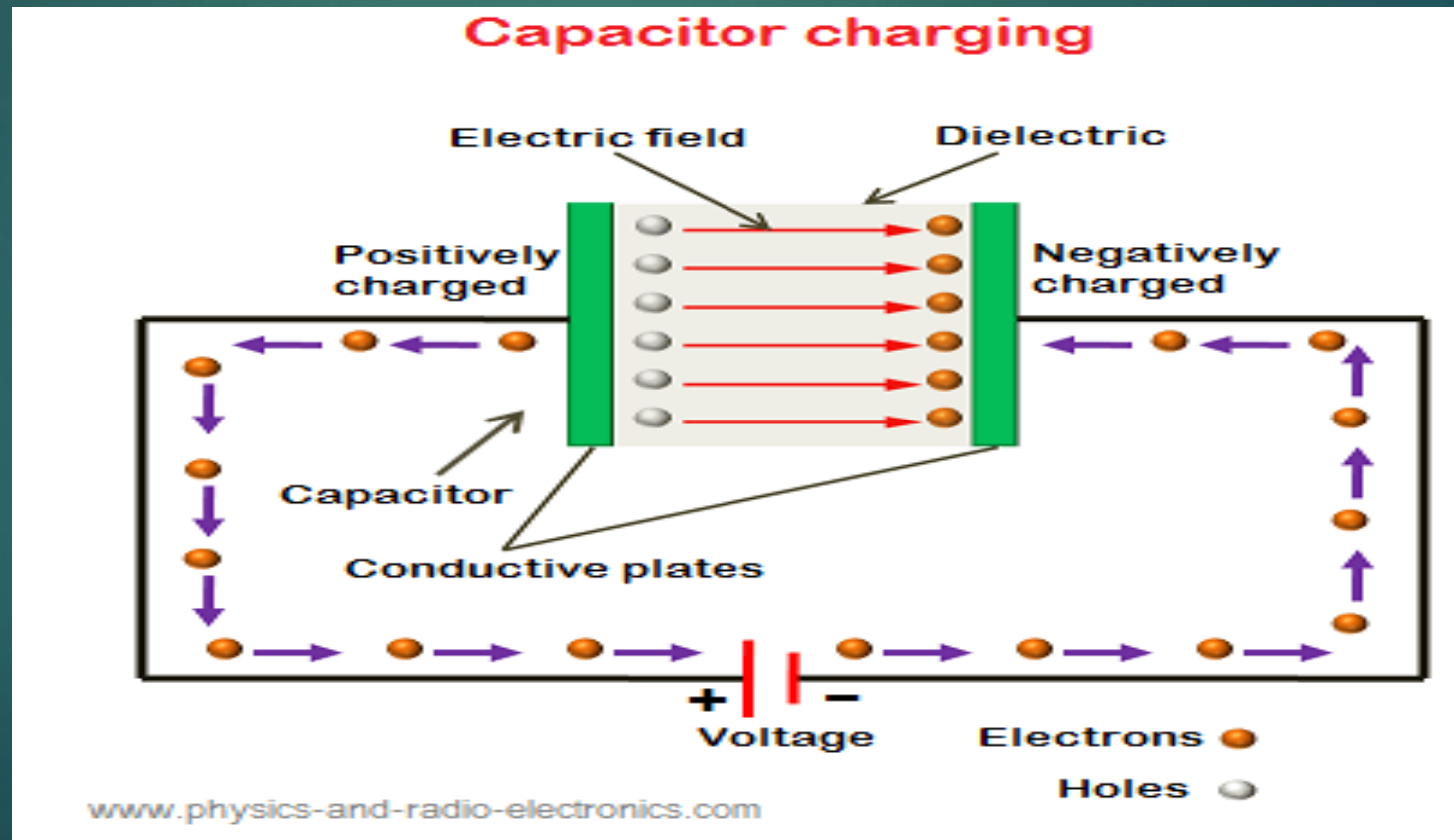


Experiment #(10)

RC-Circuit





Objective

- 1.To study the charging and discharging of a capacitor.
- 2.To find the value of a large resistor.

Apparatus

Power Supply
(liquid battery)



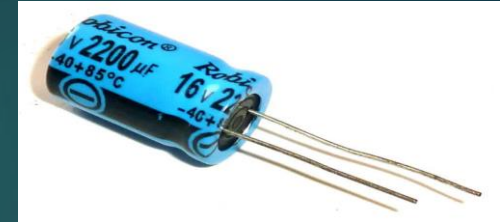
Resistance Box



Avometer



Capacitor



Banana Wires



Stop Watch



Part (1): Charging

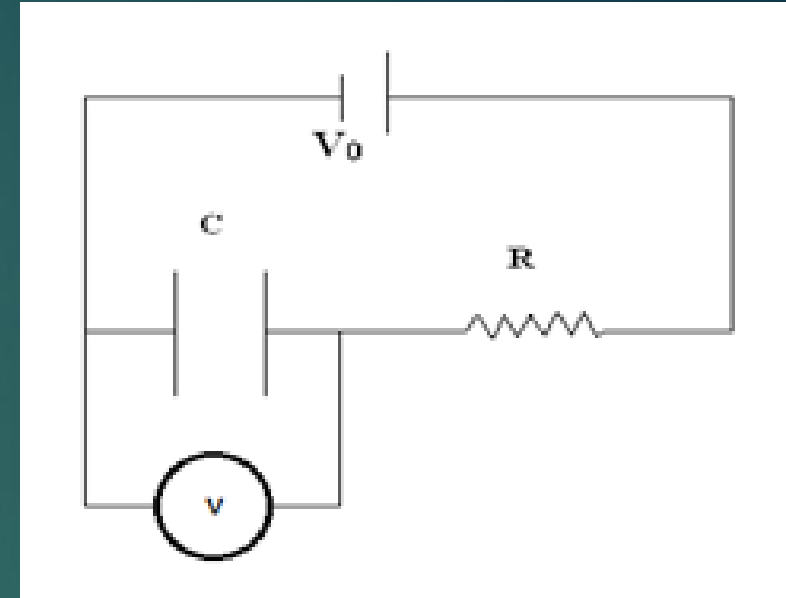
$$q(t) = Q_{max}(1 - e^{-\frac{t}{RC}}) \dots\dots\dots(1)$$

$$\frac{q(t)}{C} = \frac{Q_{max}}{C} (1 - e^{-\frac{t}{RC}}) \dots\dots\dots(2)$$

Since $V = \frac{q}{C}$

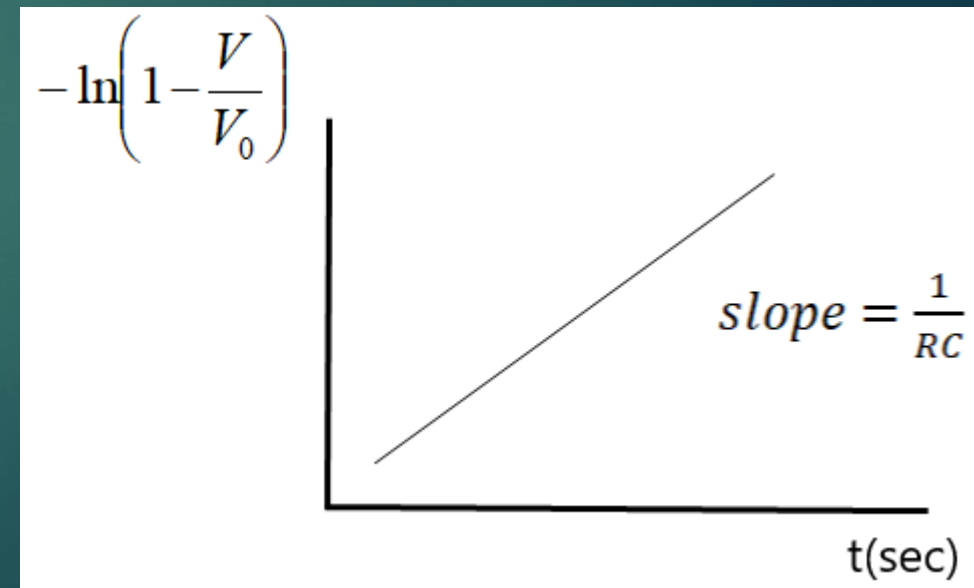
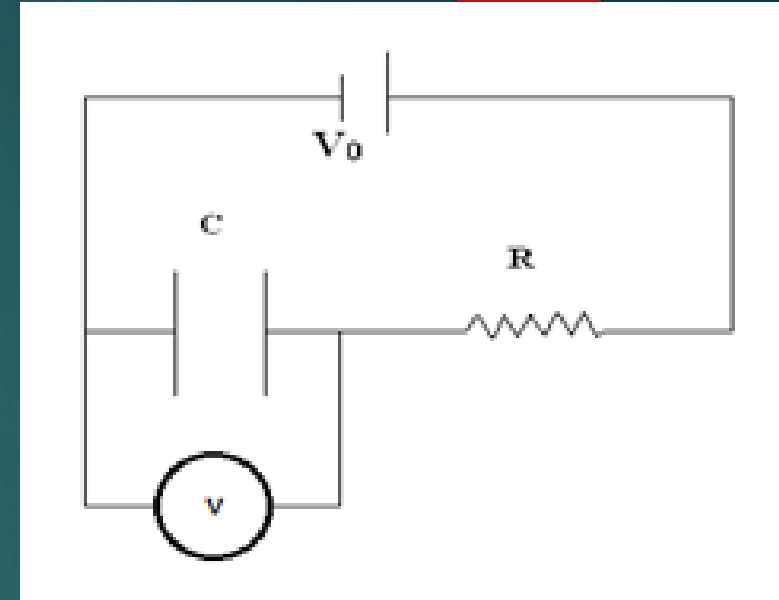
$$V(t) = V_0(1 - e^{-\frac{t}{RC}}) \dots\dots\dots(3)$$

$$- \ln \left(1 - \frac{V(t)}{V_0} \right) = \frac{t}{RC} \dots\dots\dots(4)$$



Procedure

1. Connect the circuit shown in the figure.(let one end of any wire disconnected and let $R=50K\Omega$).
2. Connect the circuit with starting the time on.
3. Record the reading of the voltmeter every 5 seconds.
4. Draw the relation between $-\ln\left(1-\frac{V}{V_0}\right)$ and the time (t).
5. From the slope calculate the capacitance C.



Part (2): Discharging

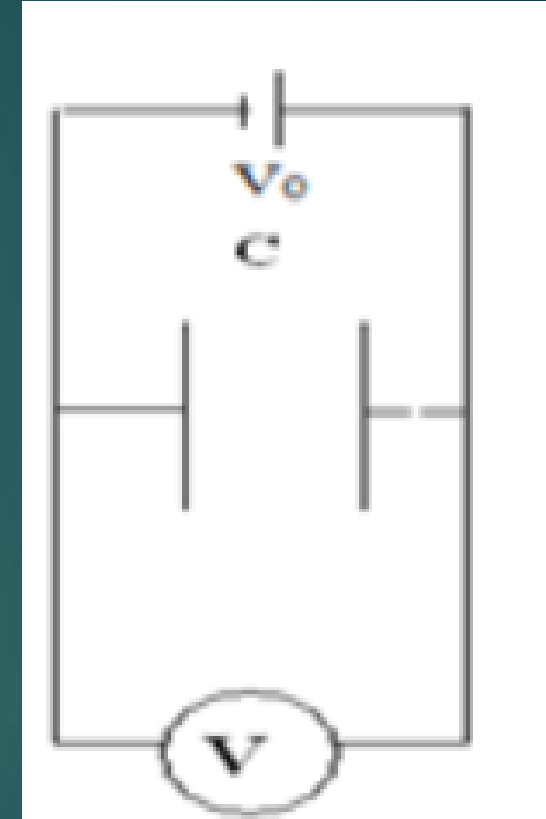
$$q(t) = Q_0 e^{-\frac{t}{R_v C}} \dots\dots\dots (1)$$

$$\frac{q(t)}{C} = \frac{Q_0}{C} e^{-\frac{t}{R_v C}} \dots\dots\dots (2)$$

Since $V = \frac{q}{C}$

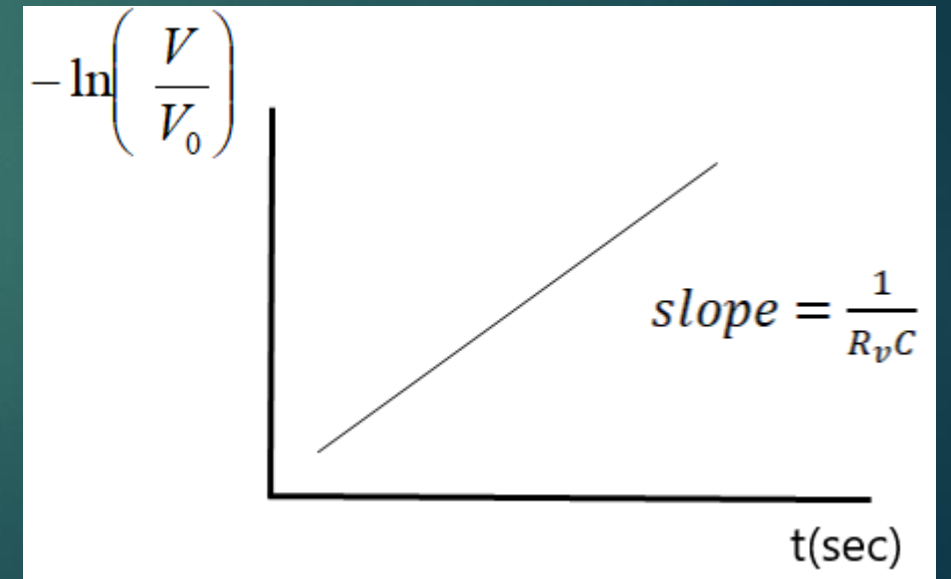
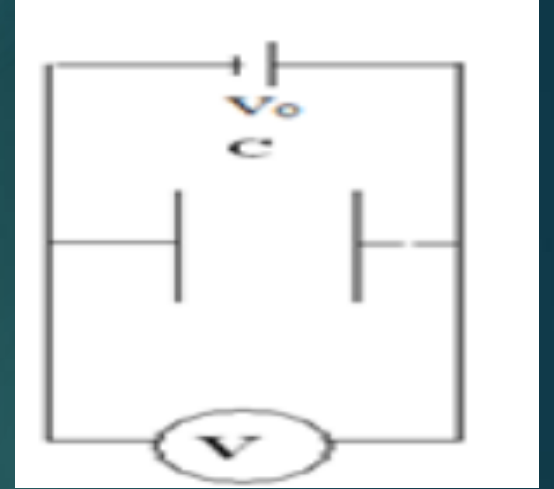
$$V(t) = V_0 e^{-\frac{t}{R_v C}} \dots\dots\dots (3)$$

$$- \ln \left(\frac{V(t)}{V_0} \right) = \frac{t}{R_v C} \dots\dots\dots (4)$$



Procedure

1. Connect the circuit shown in the figure. Disconnect the power supply with starting the time on.
2. Record the reading of the voltmeter every 10 seconds.
3. Draw the relation between $-\ln\left(\frac{V}{V_0}\right)$ and the time (t).
4. From the slope calculate the value of the voltmeter resistance R_v .





THANK YOU